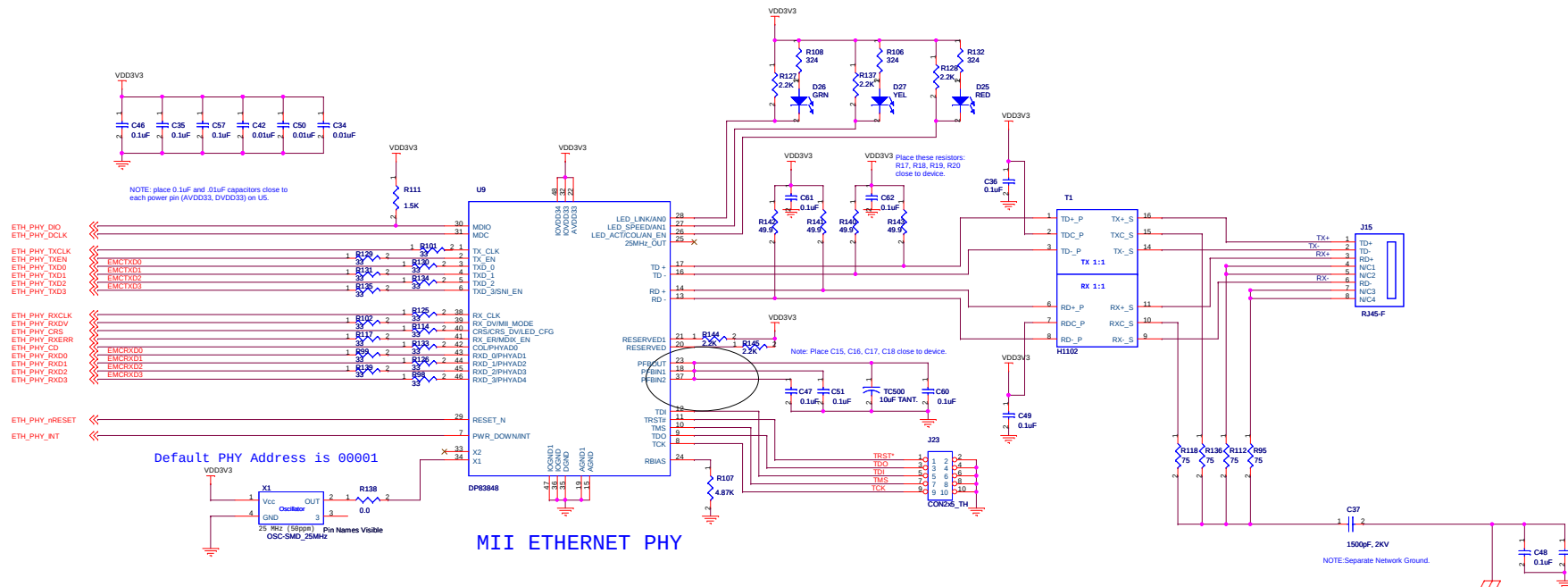
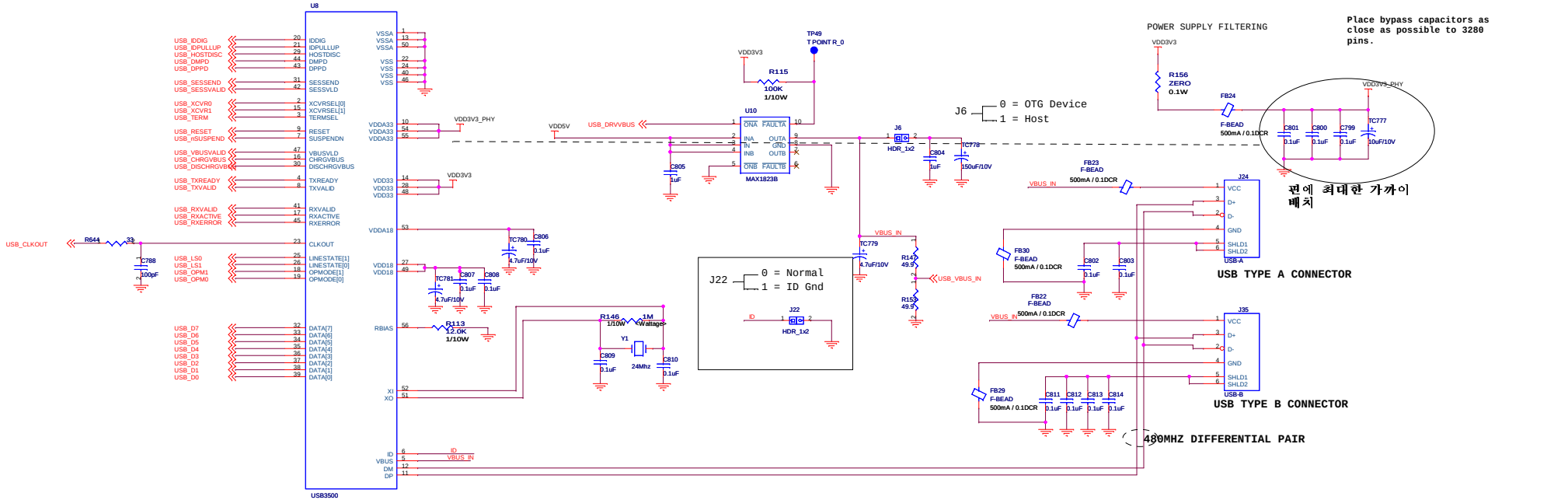
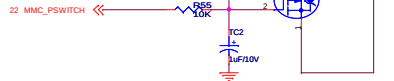
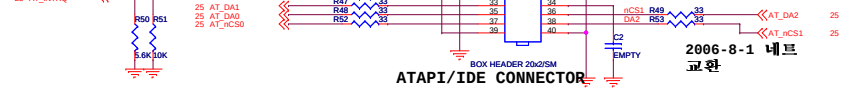
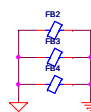
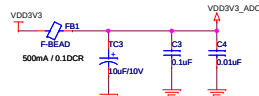
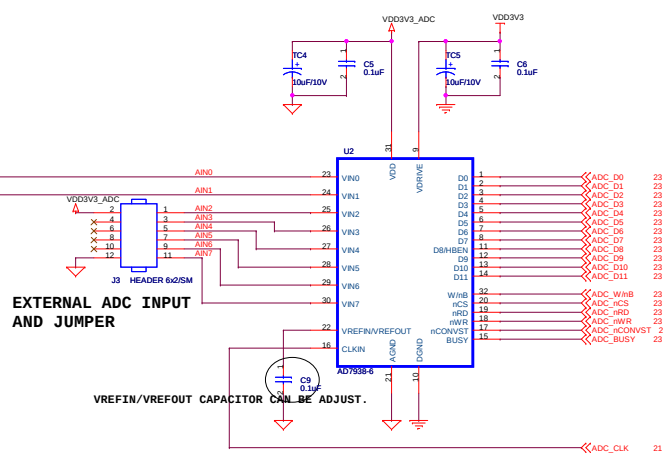
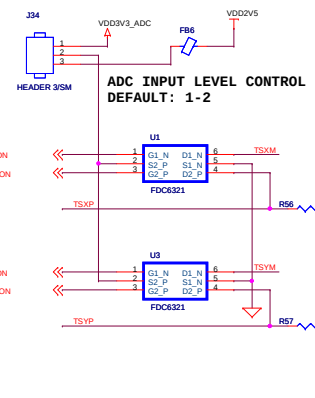
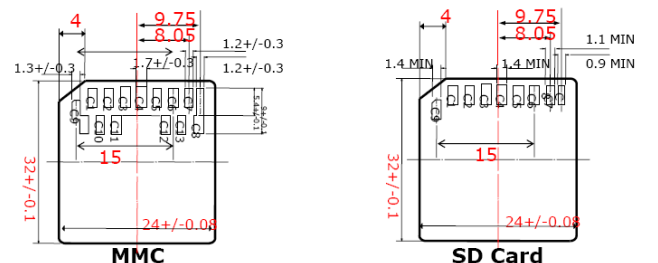
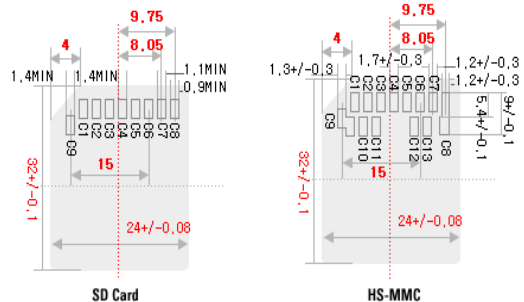
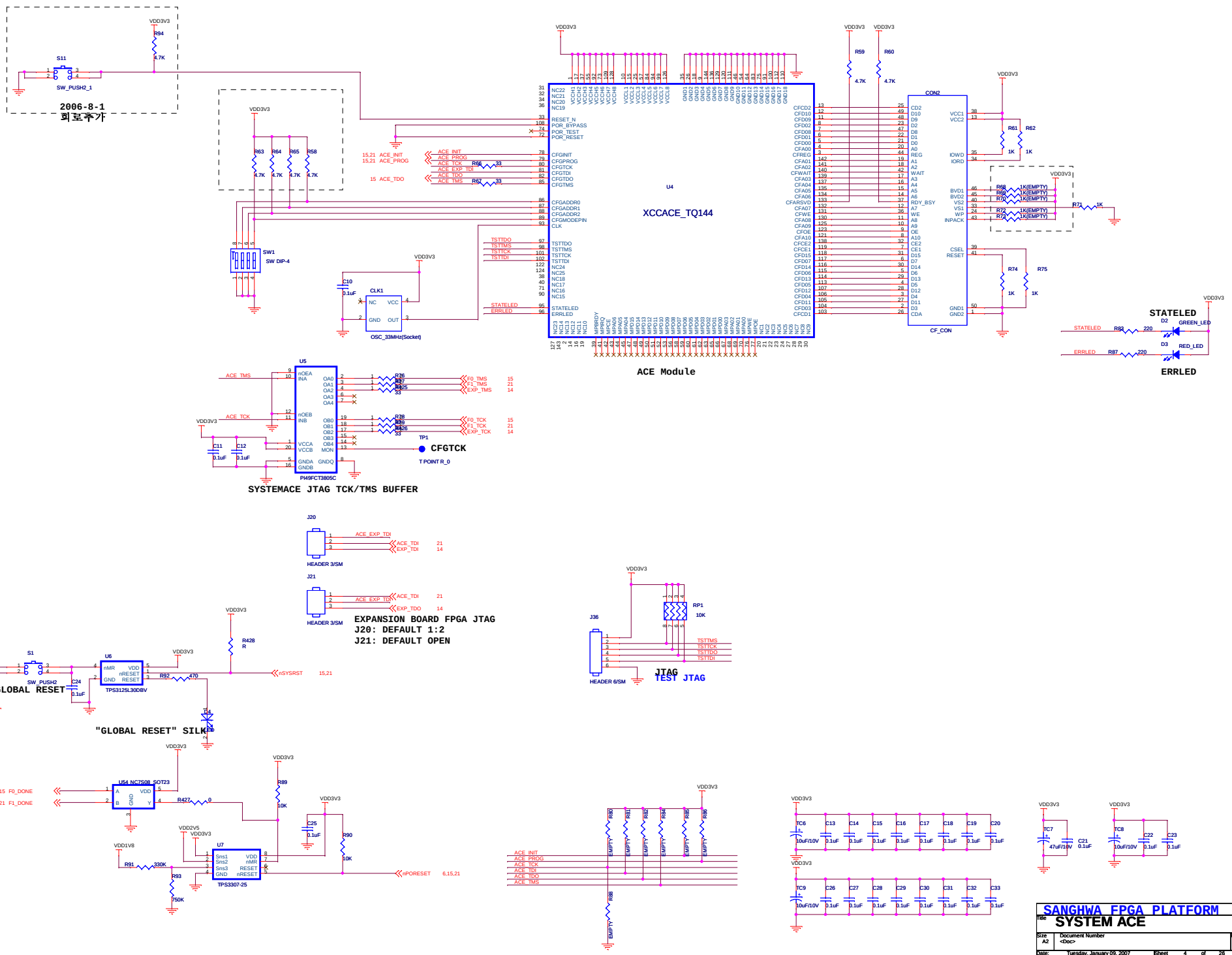


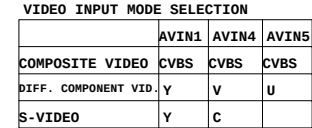


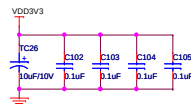
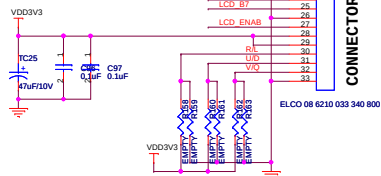


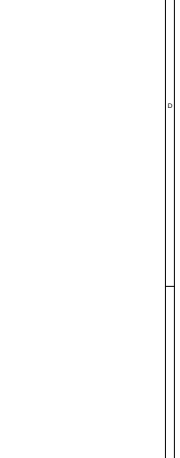
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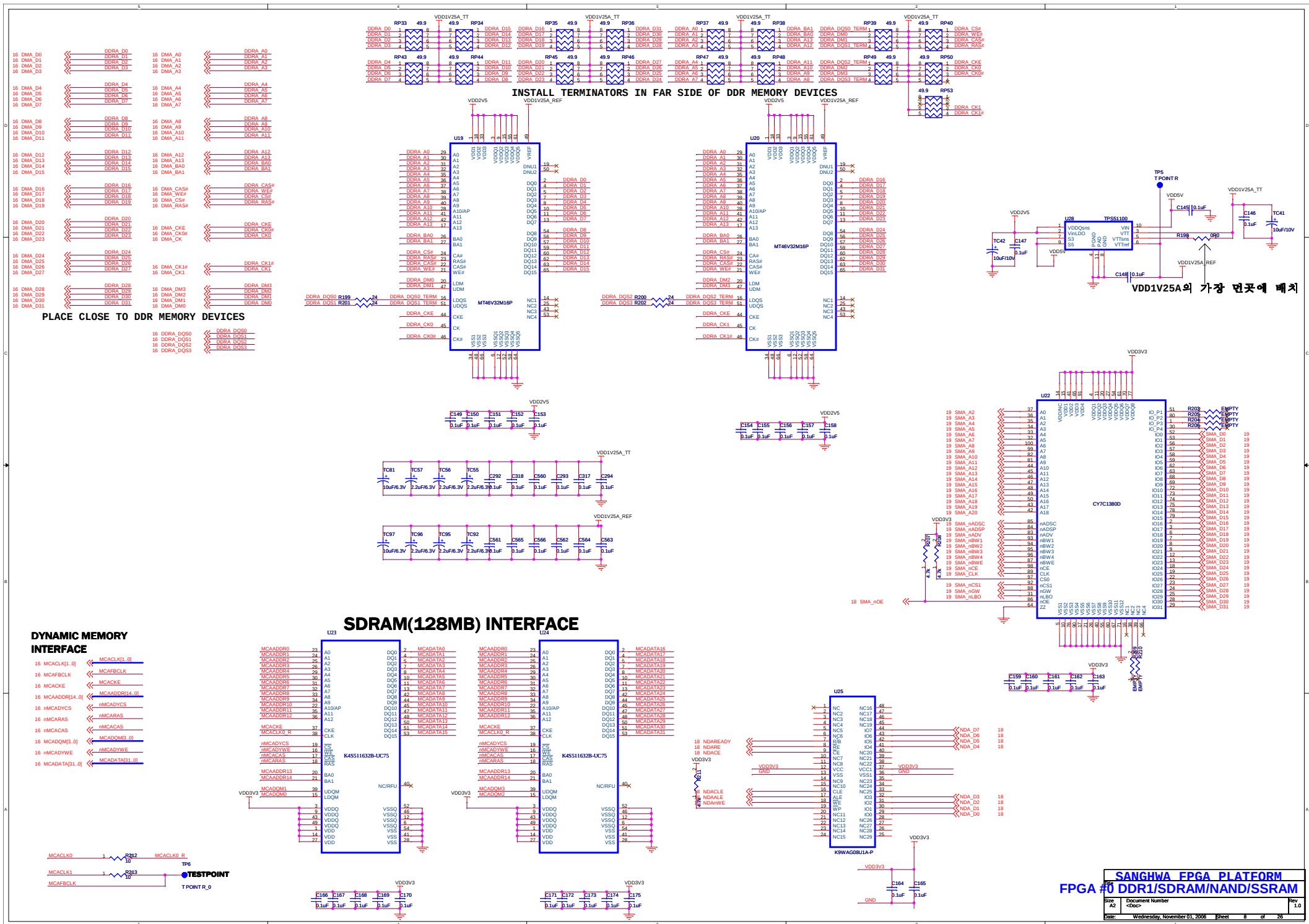


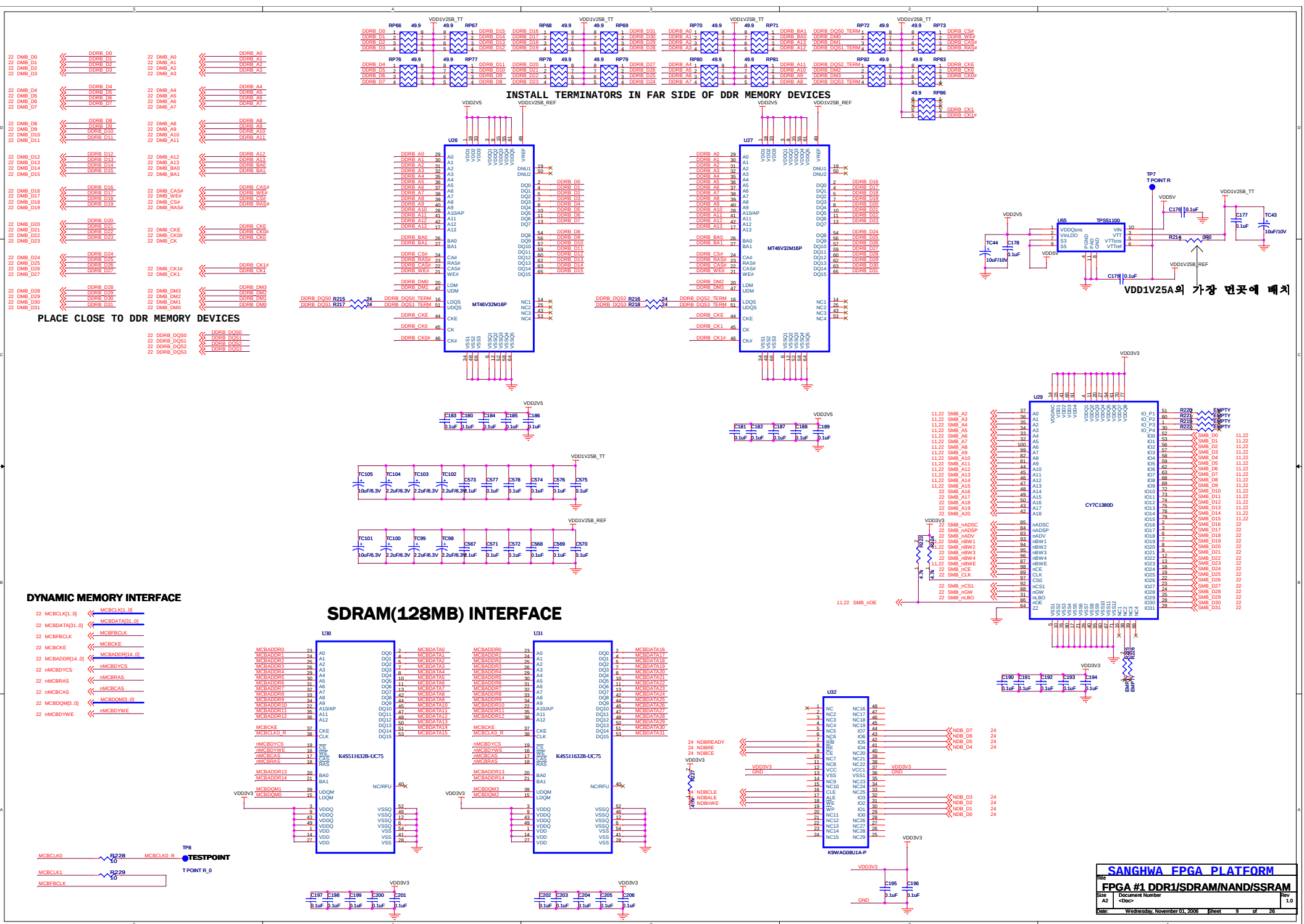






PLATFORM	
RAM	
	Rev 1.0
7 of 26	





ETH MISC SIGNALS

22 ETH_INTR
22 ETH_CS
22 ETH_RESET
22 ETH_LCLK

22 SMB_A1
9.22 SMB_A2
9.22 SMB_A3
9.22 SMB_A4
9.22 SMB_A5
9.22 SMB_A6
9.22 SMB_A7
9.22 SMB_A8
9.22 SMB_A9
9.22 SMB_A10
9.22 SMB_A11
9.22 SMB_A12
9.22 SMB_A13
9.22 SMB_A14
9.22 SMB_A15

U37B
SYSTEM Interface
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15

ETH_CS
ETH_RESET
ETH_INTR
ETH_LCLK
nBEO
nBE1
nBE2
nBE3
nDATAS
W_nR
nRCLE
nRDVRTN
nVLBUS
nADS
nLDEV
nSRDY

LAN91C111

U37A
GND-9
GND-8
GND-7
GND-6
GND-5
GND-4
GND-3
GND-2
GND-1
GND

LAN91C111

U37C
TP0p
TP0n
nCNTRL
nCSOUT
LBK
XTAL1
XTAL2
X25out
nLEDA
nLEDB
RBIAS

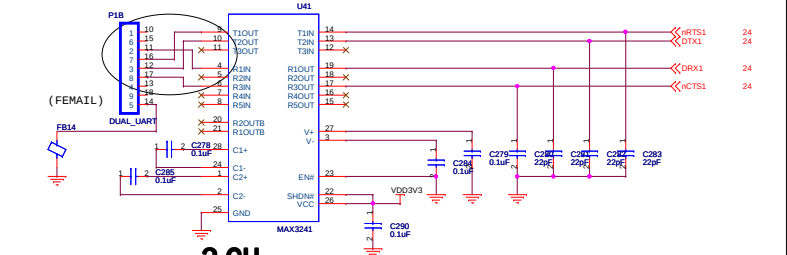
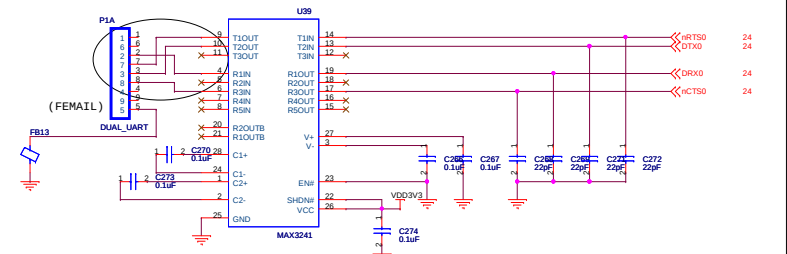
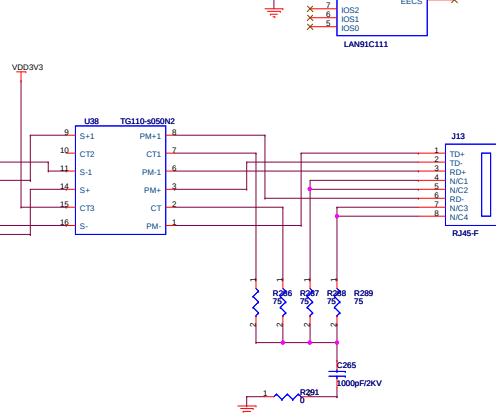
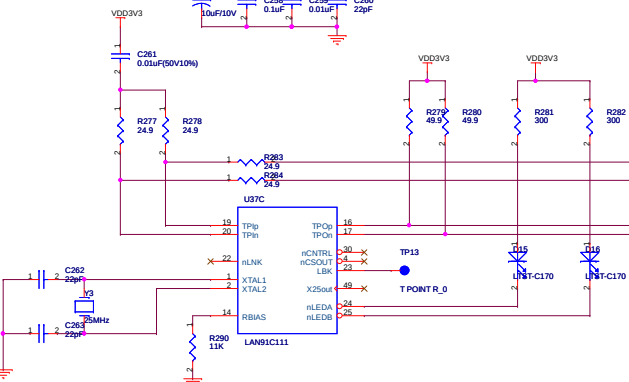
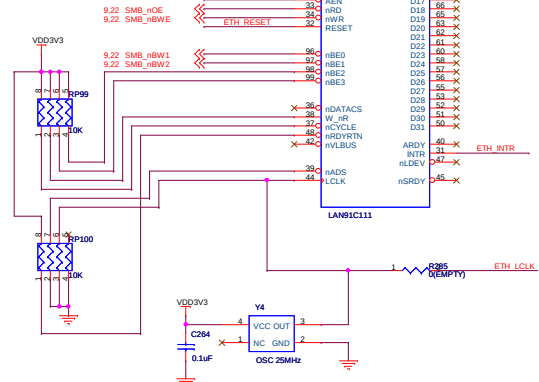
LAN91C111

U37D
RTE Interface
MDI
MDO
RXD3
RXD2
RXD1
RXD0
TXD3
TXD2
TXD1
TXD0
CRS100
COL100
RX_DV
RX_ER
TX25
TX25

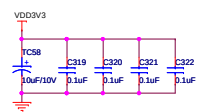
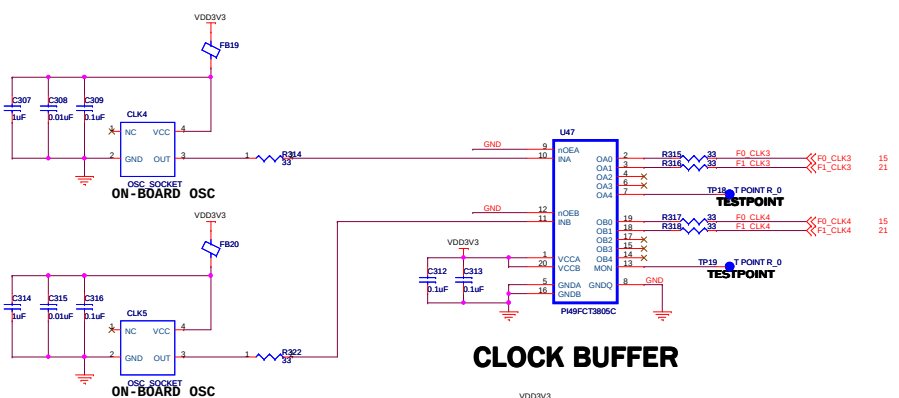
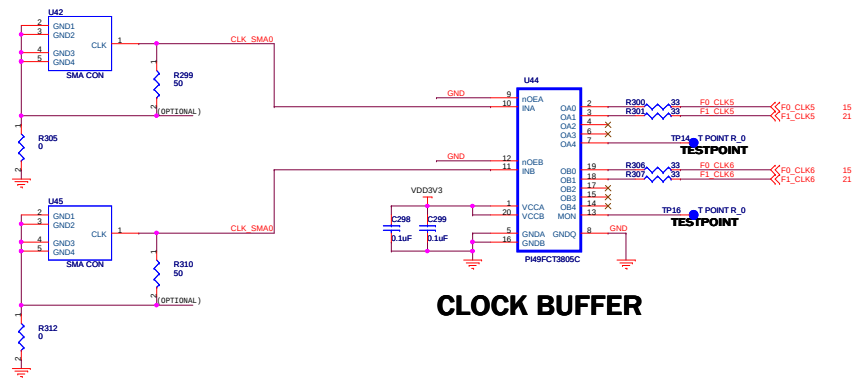
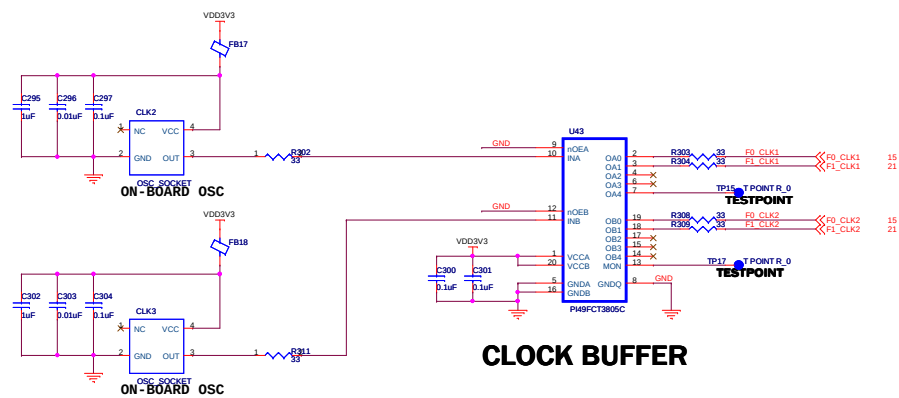
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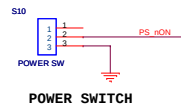
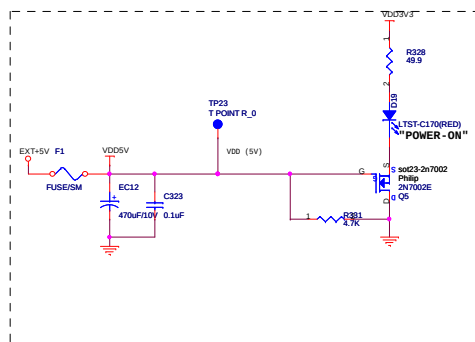
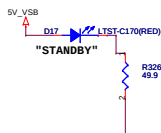
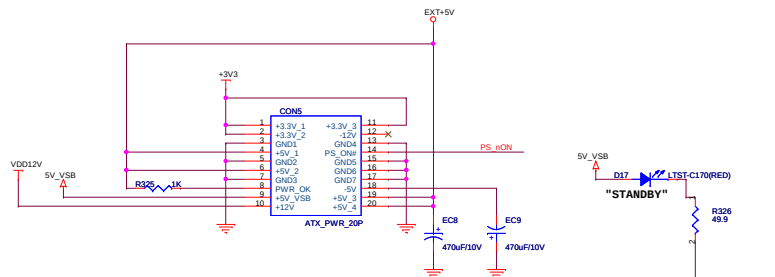
U37E
TRENK Interface
EEDI
EENEEP
IOS2
IOS1
IOS0

LAN91C111



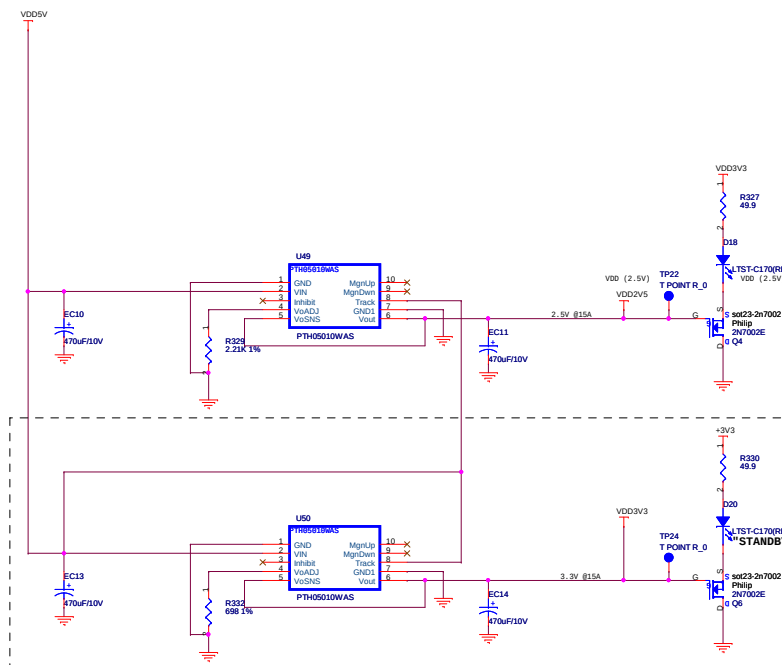
2-CH
UART



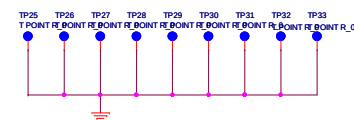


파워스위치 변경

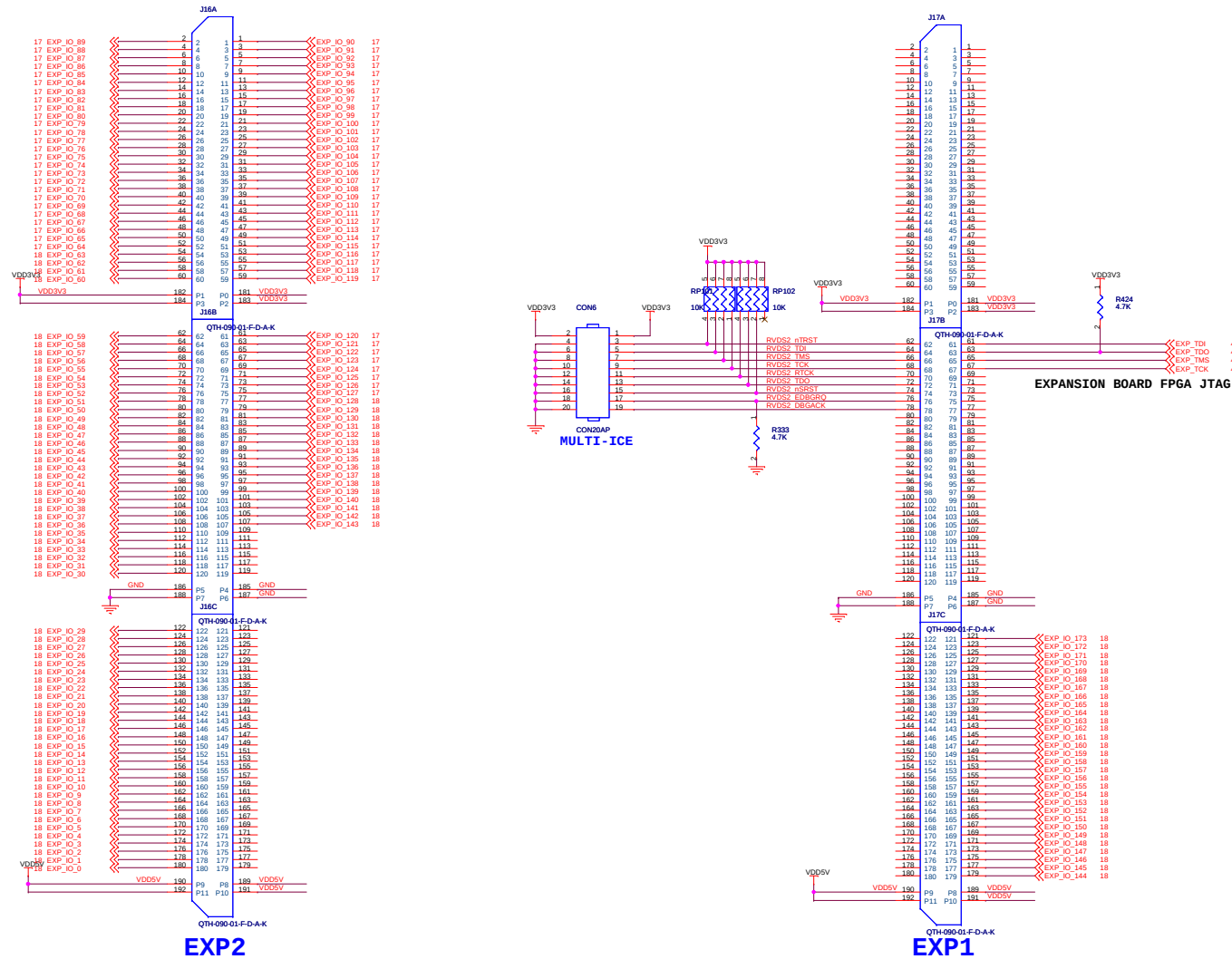
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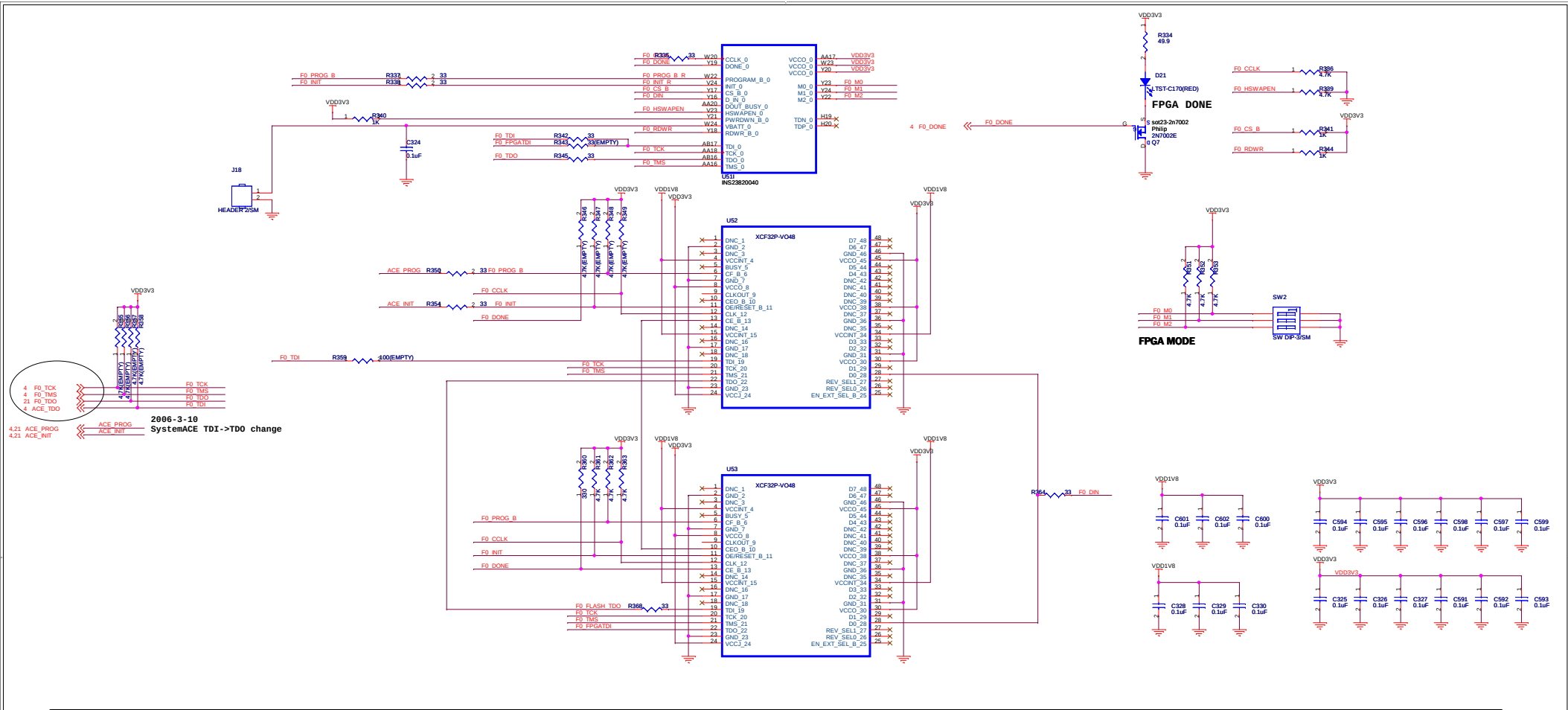


GND Point

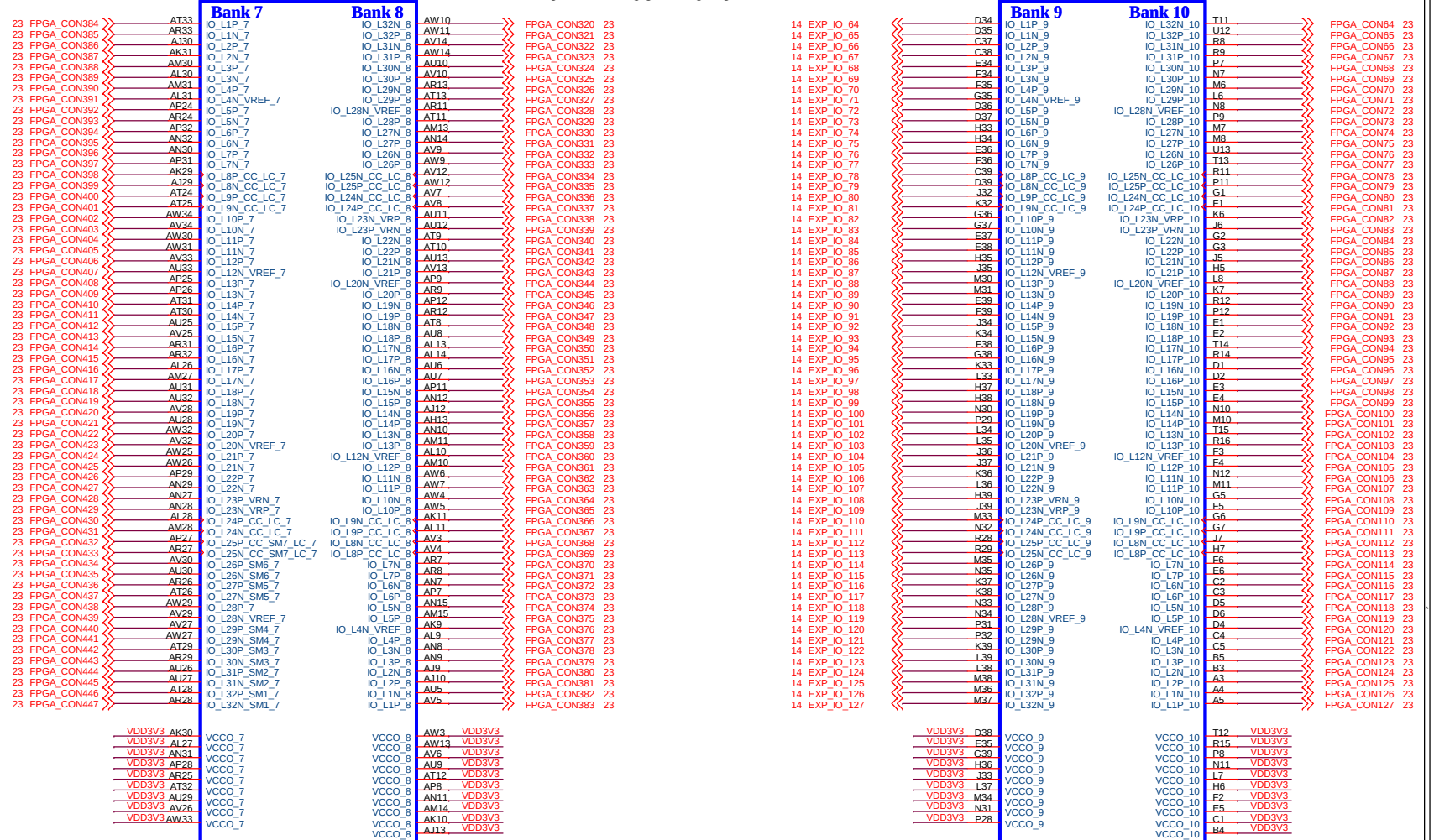


EXPANSION SLOT FOR FPGA #0



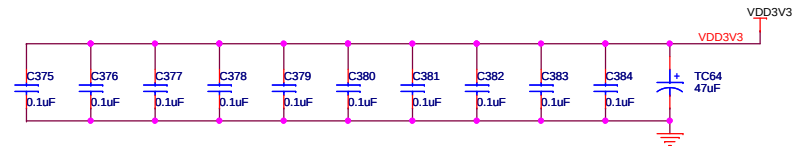
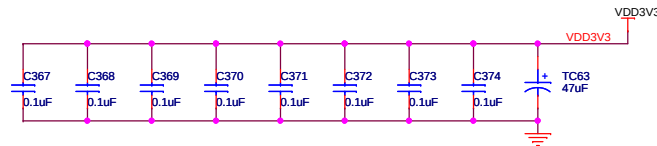


FPGA INTERCONNECTION



U51D
INS23832289

U51E
INS23834214



8 SMA_nOE

NAND FLASH

8 NDAREADY
8 NDARE
8 NDACE
8 NDACLE
8 NDAALE
8 NDAAnWE
8 NDA_D7
8 NDA_D6
8 NDA_D5
8 NDA_D4
8 NDA_D3
8 NDA_D2
8 NDA_D1
8 NDA_D0

14 EXP IO 173
14 EXP IO 172
14 EXP IO 171
14 EXP IO 169
14 EXP IO 168
14 EXP IO 167
14 EXP IO 165
14 EXP IO 164
14 EXP IO 163
14 EXP IO 161
14 EXP IO 160
14 EXP IO 159
14 EXP IO 158
14 EXP IO 157
14 EXP IO 156
14 EXP IO 155
14 EXP IO 154
14 EXP IO 152
14 EXP IO 151
14 EXP IO 150
14 EXP IO 149
14 EXP IO 148
14 EXP IO 147
14 EXP IO 146
14 EXP IO 145
14 EXP IO 144
14 EXP IO 143
14 EXP IO 142
14 EXP IO 140
14 EXP IO 139
14 EXP IO 138
14 EXP IO 137
14 EXP IO 136
14 EXP IO 135
14 EXP IO 134
14 EXP IO 133
14 EXP IO 132
14 EXP IO 131
14 EXP IO 130
14 EXP IO 129
14 EXP IO 128

VDD3V3	AD28
VDD3V3	AE25
VDD3V3	AF32
VDD3V3	AG29
VDD3V3	AJ33
VDD3V3	AM34
VDD3V3	AP38
VDD3V3	AR35
VDD3V3	AU39
VDD3V3	AV36

VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11

U51F
INS23839244

Bank 12

IO_L32P_L2
IO_L31N_L2
IO_L31P_L2
IO_L30N_L2
IO_L30P_L2
IO_L29N_L2
IO_L29P_L2
IO_L28N_VREF_L2
IO_L28P_L2
IO_L27N_L2
IO_L27P_L2
IO_L26N_L2
IO_L26P_L2
IO_L25N_CC_LC_L2
IO_L25P_CC_LC_L2
IO_L24N_CC_LC_L2
IO_L24P_CC_LC_L2
IO_L23P_VRN_L2
IO_L22N_L2
IO_L22P_L2
IO_L21N_L2
IO_L21P_L2
IO_L20N_VREF_L2
IO_L20P_L2
IO_L19N_L2
IO_L19P_L2
IO_L18N_L2
IO_L18P_L2
IO_L17N_L2
IO_L17P_L2
IO_L16N_L2
IO_L16P_L2
IO_L15N_L2
IO_L15P_L2
IO_L14N_L2
IO_L14P_L2
IO_L13N_L2
IO_L13P_L2
IO_L12N_VREF_L2
IO_L12P_L2
IO_L11N_L2
IO_L11P_L2
IO_L10N_L2
IO_L10P_L2
IO_L9N_CC_LC_L2
IO_L9P_CC_LC_L2
IO_L8N_CC_LC_L2
IO_L8P_CC_LC_L2
IO_L7N_L2
IO_L7P_L2
IO_L6N_L2
IO_L6P_L2
IO_L5N_L2
IO_L5P_L2
IO_L4N_VREF_L2
IO_L4P_L2
IO_L3N_L2
IO_L3P_L2
IO_L2N_L2
IO_L2P_L2
IO_L1N_L2
IO_L1P_L2

VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12

AT2	VDD3V3
AR5	VDD3V3
AN1	VDD3V3
AM4	VDD3V3
AL7	VDD3V3
AJ3	VDD3V3
AH6	VDD3V3
AG9	VDD3V3
AF12	VDD3V3

AU8
AT6
AP4
AP5
AP6
AR5
AM6
AM7
AR4
AT4
AP2
AR3
AU1
AU2
AR1
AT1
AK8
AL8
AL5
AM5
AN4
AN5
AK7
AH9
AH10
AN2
AN3
AK6
AL6
AK4
AL4
AP1
AP2
AM1
AM2
AL3
AM3
AE11
AE12
AG10
AE11
AK2
AK3
AJ5
AJ6
AH4
AJ4
AH7
AG8
AF8
AF9
AK1
AL1
AJ1
AJ2
AG6
AG7
AH2
AH3
AG5
AH5

AT2	VDD3V3
AR5	VDD3V3
AN1	VDD3V3
AM4	VDD3V3
AL7	VDD3V3
AJ3	VDD3V3
AH6	VDD3V3
AG9	VDD3V3
AF12	VDD3V3

14 EXP IO 1
14 EXP IO 2
14 EXP IO 3
14 EXP IO 4
14 EXP IO 5
14 EXP IO 6
14 EXP IO 7
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14 EXP IO 9
14 EXP IO 10
14 EXP IO 11
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14 EXP IO 54
14 EXP IO 55
14 EXP IO 56
14 EXP IO 57
14 EXP IO 58
14 EXP IO 59
14 EXP IO 60
14 EXP IO 61
14 EXP IO 62

VDD3V3	AA37
VDD3V3	P38
VDD3V3	R35
VDD3V3	T32
VDD3V3	U29
VDD3V3	U39
VDD3V3	V26
VDD3V3	V36
VDD3V3	W33

VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13

U51G
INS23841562

IO_L1N_13
IO_L2P_13
IO_L2N_13
IO_L3N_13
IO_L3P_13
IO_L4P_13
IO_L4N_VREF_13
IO_L5P_13
IO_L5N_13
IO_L6P_13
IO_L6N_13
IO_L7P_13
IO_L7N_13
IO_L8P_CC_LC_13
IO_L9P_CC_LC_13
IO_L9N_CC_LC_13
IO_L10P_13
IO_L10N_13
IO_L11P_13
IO_L11N_13
IO_L12P_13
IO_L12N_VREF_13
IO_L13P_13
IO_L13N_13
IO_L14P_13
IO_L14N_13
IO_L15P_13
IO_L15N_13
IO_L16P_13
IO_L16N_13
IO_L17P_13
IO_L17N_13
IO_L18P_13
IO_L18N_13
IO_L19P_13
IO_L19N_13
IO_L20P_13
IO_L20N_VREF_13
IO_L21P_13
IO_L21N_13
IO_L22P_13
IO_L22N_13
IO_L23P_VRN_13
IO_L23N_VRP_13
IO_L24P_CC_LC_13
IO_L24N_CC_LC_13
IO_L25P_CC_LC_13
IO_L25N_CC_LC_13
IO_L26P_13
IO_L26N_13
IO_L27P_13
IO_L27N_13
IO_L28P_13
IO_L28N_VREF_13
IO_L29P_13
IO_L29N_13
IO_L30P_13
IO_L30N_13
IO_L31P_13
IO_L31N_13
IO_L32P_13
IO_L32N_13

VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13
VCCO_13

U51G
INS23841562

Bank 14

IO_132P_14
IO_131N_14
IO_131P_14
IO_129N_14
IO_130P_14
IO_129N_14
IO_129P_14
IO_128N_VREF_14
IO_128P_14
IO_127N_14
IO_127P_14
IO_126N_14
IO_126P_14
IO_125N_VREF_14
IO_125P_CC_LC_14
IO_124N_CC_LC_14
IO_124P_CC_LC_14
IO_123P_VRN_14
IO_122N_14
IO_122P_14
IO_121N_14
IO_121P_14
IO_120N_VREF_14
IO_120P_14
IO_119N_14
IO_119P_14
IO_118N_14
IO_118P_14
IO_117N_14
IO_117P_14
IO_116N_14
IO_116P_14
IO_115N_14
IO_115P_14
IO_114N_14
IO_114P_14
IO_113N_14
IO_113P_14
IO_112N_VREF_14
IO_112P_14
IO_111N_14
IO_111P_14
IO_110N_14
IO_110P_14
IO_109N_VREF_14
IO_109P_CC_LC_14
IO_108N_VREF_14
IO_108P_CC_LC_14
IO_107N_14
IO_107P_14
IO_106N_14
IO_106P_14
IO_105N_14
IO_105P_14
IO_104N_VREF_14
IO_104P_14
IO_103N_14
IO_103P_14
IO_102N_14
IO_102P_14
IO_101N_14
IO_LIN_14

VCCO_14
VCCO_14
VCCO_14
VCCO_14
VCCO_14
VCCO_14
VCCO_14
VCCO_14

W12

14	V13
14	A11
14	Aa11
14	Y9
14	Aa10
14	W5
14	W6
14	W9
14	W10
14	W7
14	V7
14	U2
14	J2
14	T1
14	U1
14	V5
14	U5
14	V9
14	V10
14	U6
14	U7
14	T3
14	T4
14	R1
14	R2
14	T5
14	T6
14	R3
14	R4
14	U9
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14	P1
14	P2
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14	T8
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14	N5
14	M2
14	L3
14	M3
14	K1
14	M5
14	L5
14	J1
14	J2
14	J3
14	K3
14	J4
14	H4

14	Y10	VDD3V
14	W13	VDD3V
14	V6	VDD3V
14	U9	VDD3V
14	T2	VDD3V
14	R5	VDD3V
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14	M4	VDD3V
14	J3	VDD3V

SANGHWA FPGA PLATFORM
FPGA0 BANK 11/12/13/14

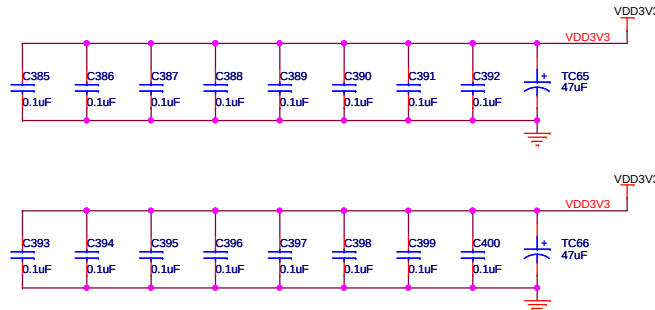
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Date: Wednesday, November 01, 2006 Sheet 18 of 26

3V



VIRTEX-4 FPGA
POWER GROUND

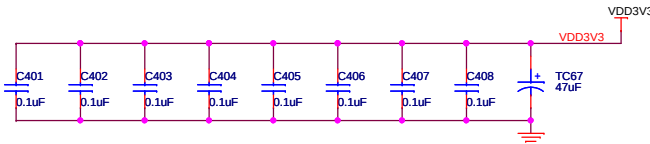
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GND AD13	GND	GND AW28	GND
GND AF13	GND	GND B29	GND
GND AK13	GND	GND M29	GND
GND AP13	GND	GND AB29	GND
GND G14	GND	GND AM29	GND
GND L14	GND	GND E30	GND
GND N14	GND	GND R30	GND
GND U14	GND	GND AE30	GND
GND W14	GND	GND AR30	GND
GND AE14	GND	GND H31	GND
GND AG14	GND	GND V31	GND
GND AJ14	GND	GND AH31	GND
GND AU14	GND	GND AV31	GND
GND K15	GND	GND A32	GND
GND M15	GND	GND L32	GND
GND P15	GND	GND AA32	GND
GND Y15	GND	GND AL32	GND
GND Y15	GND	GND D33	GND
GND AD15	GND	GND P33	GND
GND AF15	GND	GND AD33	GND
GND AK15	GND	GND AP33	GND
GND C16	GND	GND G34	GND
GND M16	GND	GND U34	GND
GND U16	GND	GND AG34	GND
GND AC16	GND	GND AL34	GND
GND AE16	GND	GND K35	GND
GND AN16	GND	GND Y35	GND
GND F17	GND	GND AK35	GND
GND T17	GND	GND C36	GND
GND V17	GND	GND N36	GND
GND AF17	GND	GND AC36	GND
GND AT17	GND	GND AN36	GND
GND W18	GND	GND F37	GND
GND AC18	GND	GND T37	GND
GND AJ18	GND	GND AE37	GND
GND AW18	GND	GND AT37	GND
GND B19	GND	GND A38	GND
GND M19	GND	GND J38	GND
GND V19	GND	GND W38	GND
GND AB19	GND	GND AJ38	GND
GND AD19	GND	GND AW38	GND
GND AM19	GND	GND B39	GND
GND E20	GND	GND M39	GND
GND R20	GND	GND AE39	GND
		GND AM39	GND
		GND AV39	GND
		GND	
GND AE20	GND	GND B1	GND
GND H21	GND	GND H1	GND
GND T21	GND	GND V1	GND
GND V21	GND	GND AH1	GND
GND AB21	GND	GND AV1	GND
GND AD21	GND	GND A2	GND
GND A22	GND	GND L2	GND
GND A22	GND	GND AL2	GND
GND U22	GND	GND AW2	GND
GND AA22	GND	GND D3	GND
GND AL22	GND	GND P3	GND
GND D23	GND	GND AD3	GND
GND P23	GND	GND AP3	GND
GND AB23	GND	GND G4	GND
GND AD23	GND	GND U4	GND
GND AE23	GND	GND AG4	GND
GND G24	GND	GND AL4	GND
GND R24	GND	GND K5	GND
GND U24	GND	GND Y5	GND
GND AA24	GND	GND AK5	GND
GND AC24	GND	GND C6	GND
GND AG24	GND	GND N6	GND
GND AU24	GND	GND AC6	GND
GND K25	GND	GND AN6	GND
GND P25	GND	GND F7	GND
GND Y25	GND	GND T7	GND
GND AB25	GND	GND AF7	GND
GND AE25	GND	GND AT7	GND
GND AH25	GND	GND J8	GND
GND AK25	GND	GND W8	GND
GND C26	GND	GND AJ8	GND
GND L26	GND	GND AW8	GND
GND N26	GND	GND B9	GND
GND R26	GND	GND M9	GND
GND AA26	GND	GND AB9	GND
GND AC26	GND	GND AM9	GND
GND AG26	GND	GND E10	GND
GND AJ26	GND	GND R10	GND
GND AN26	GND	GND AE10	GND
GND F27	GND	GND AR10	GND
GND K27	GND	GND H11	GND
GND P27	GND	GND V11	GND
GND T27	GND	GND AH11	GND
GND AF27	GND	GND AV11	GND
GND AH27	GND	GND A12	GND
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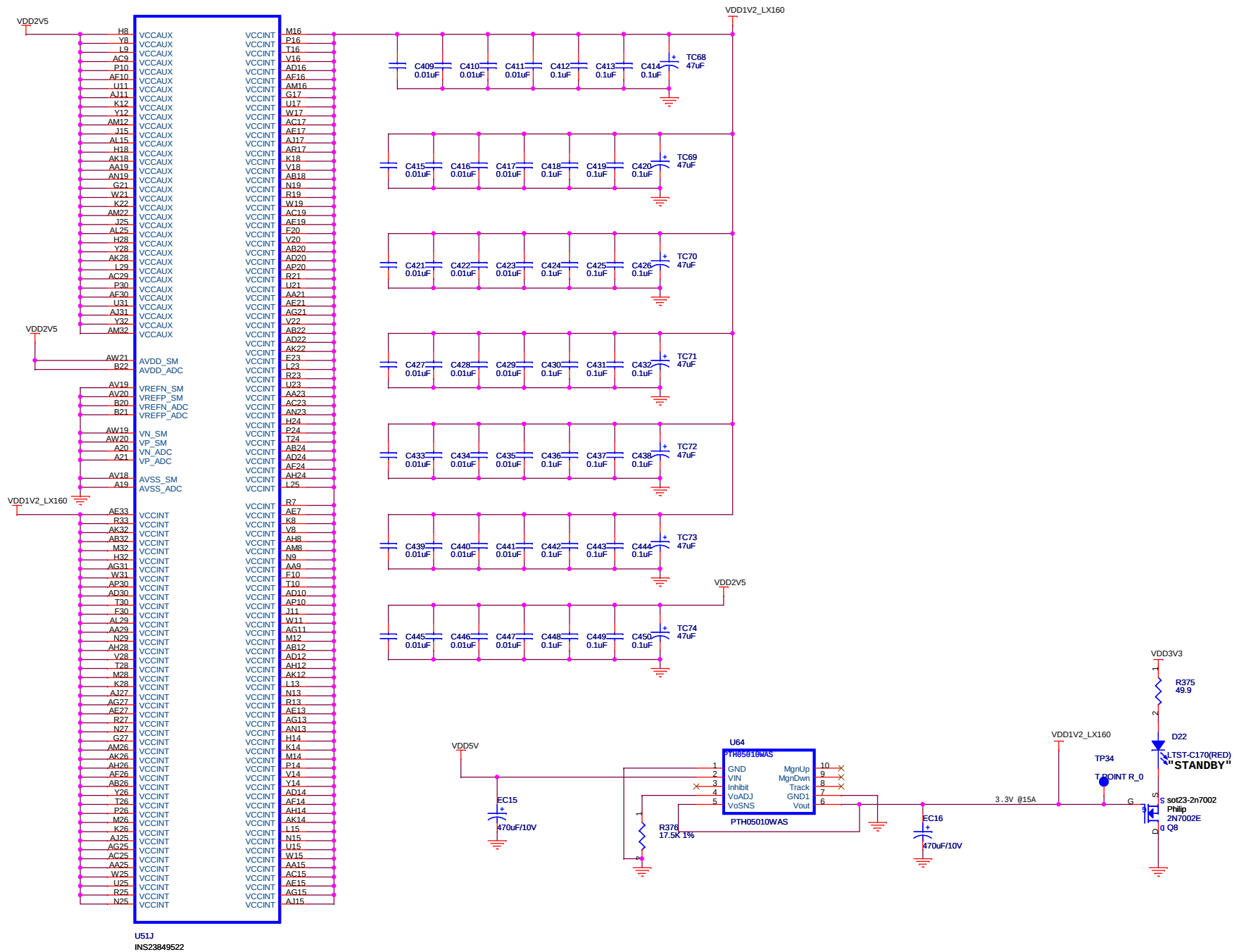
SSRAM INTERFACE

8 SMA_A2	Y27
8 SMA_A3	A28
8 SMA_A4	W29
8 SMA_A5	W30
8 SMA_A6	AA34
8 SMA_A7	AA35
8 SMA_A8	Y29
8 SMA_A9	AA30
8 SMA_A10	AC38
8 SMA_A11	AC39
8 SMA_A12	Y31
8 SMA_A13	AC35
8 SMA_A14	AB35
8 SMA_A15	AB33
8 SMA_A16	AB33
8 SMA_A17	AA33
8 SMA_A18	AC33
8 SMA_A19	AD37
8 SMA_A20	AC37
8 SMA_nADSP	AC32
8 SMA_nADV	AB31
8 SMA_nBW1	AD39
8 SMA_nBW2	AE38
8 SMA_nBW3	AE39
8 SMA_nBW4	AD35
8 SMA_nBWE	AD36
8 SMA_nCE	AC30
8 SMA_CLK	AB30
8 SMA_nCS1	AE37
8 SMA_nCW	AE38
8 SMA_nLBO	AB27
8 SMA_D31	AB28
8 SMA_D30	AE34
8 SMA_D29	AD34
8 SMA_D28	AD31
8 SMA_D27	AD32
8 SMA_D26	AE36
8 SMA_D25	AE36
8 SMA_D24	AJ39
8 SMA_D23	AH39
8 SMA_D22	AG37
8 SMA_D21	AG38
8 SMA_D20	AH37
8 SMA_D19	AH38
8 SMA_D18	AE34
8 SMA_D17	AE35
8 SMA_D16	AJ36
8 SMA_D15	AJ37
8 SMA_D14	AG35
8 SMA_D13	AG36
8 SMA_D12	AJ35
8 SMA_D11	AH35
8 SMA_D10	AH35
8 SMA_D9	AK39
8 SMA_D8	AM37
8 SMA_D7	AM38
8 SMA_D6	AL39
8 SMA_D5	AM36
8 SMA_D4	AL36
8 SMA_D3	AK36
8 SMA_D2	AK37
8 SMA_D1	AK37
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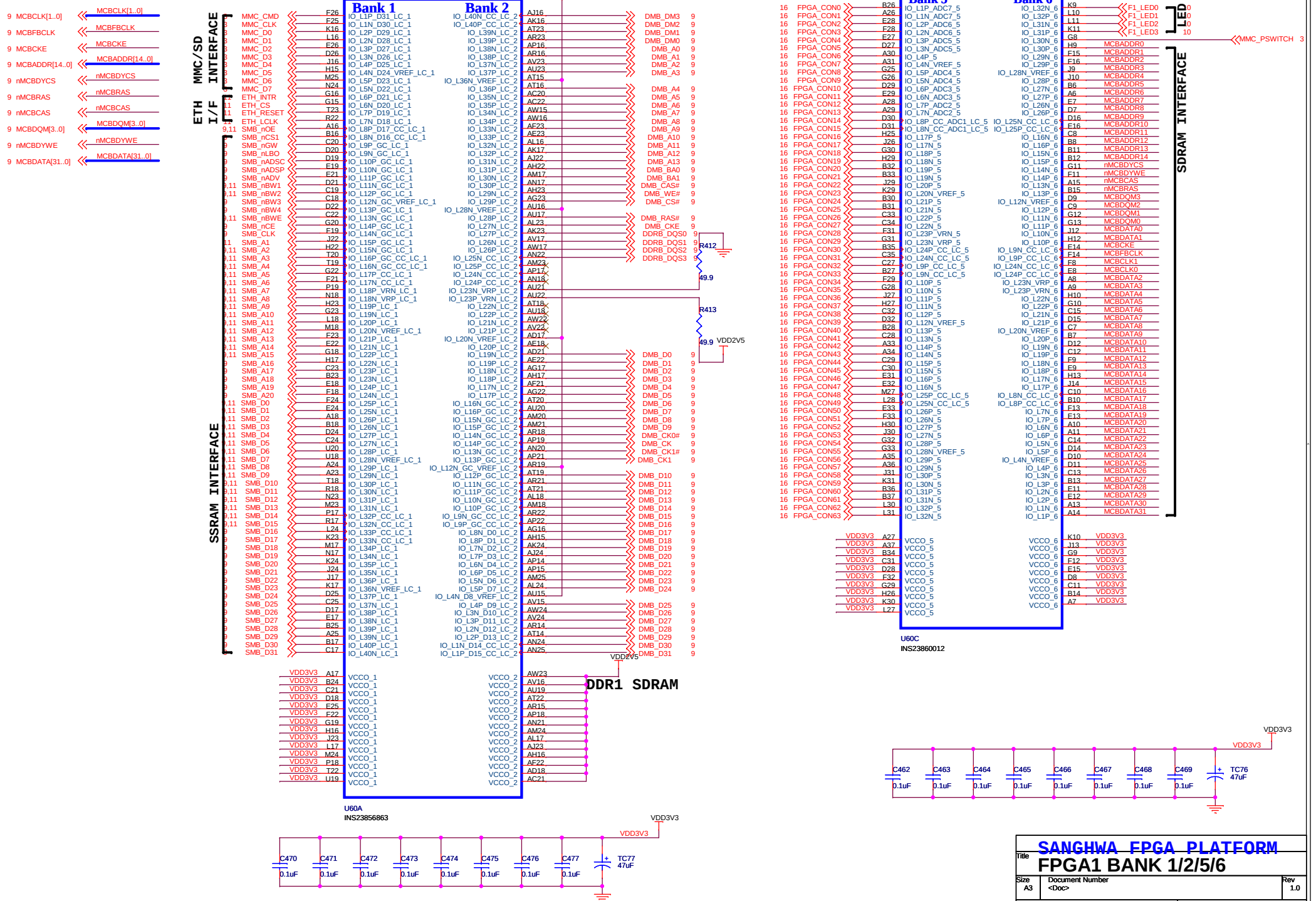
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VDD3V3	AE35	VCCO_15	
VDD3V3	AG39	VCCO_15	
VDD3V3	AH36	VCCO_15	
VDD3V3	AL37	VCCO_15	
VDD3V3	Y30	VCCO_15	

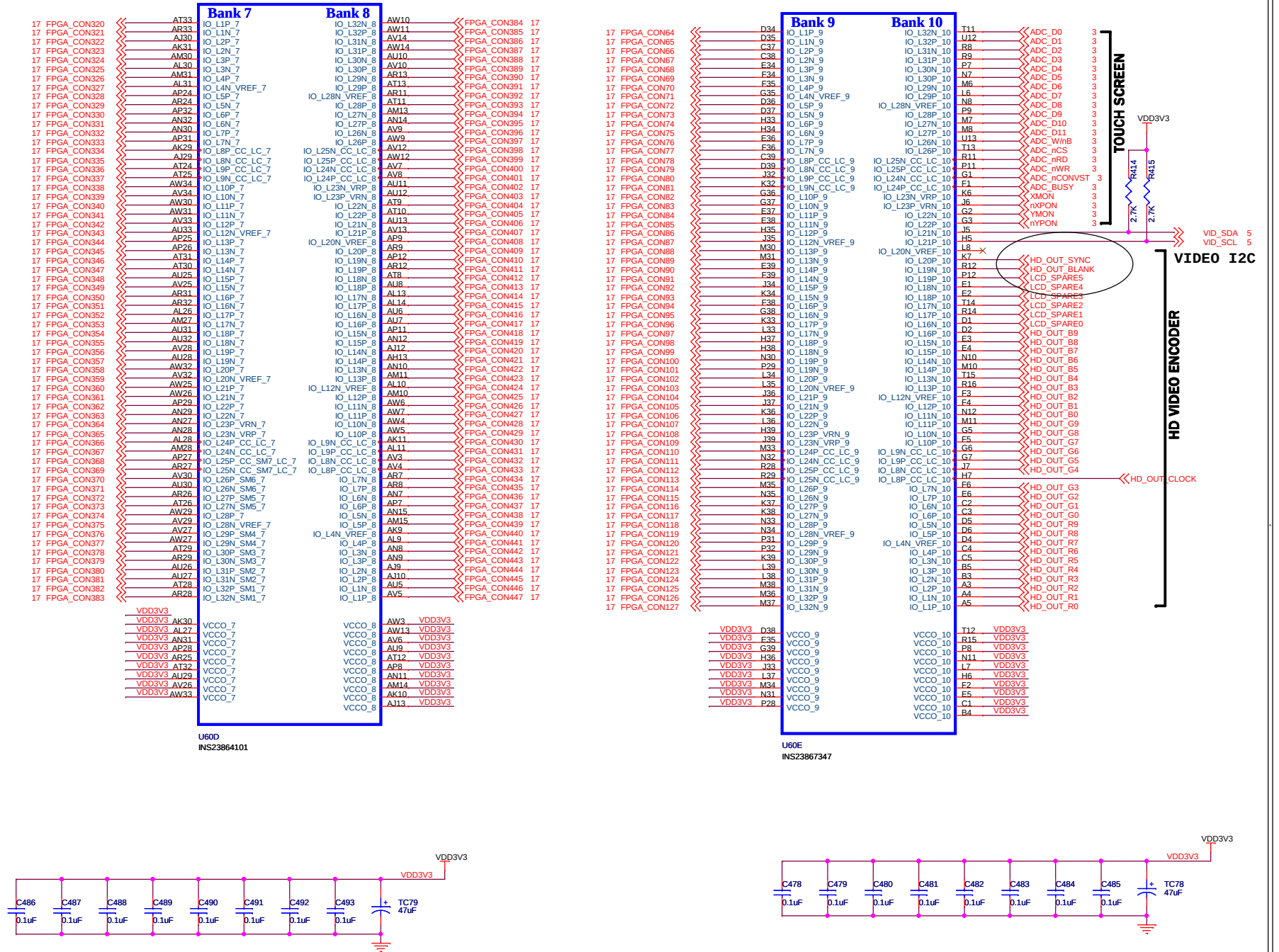
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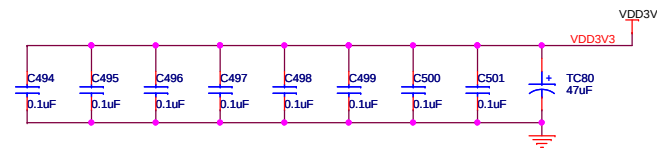
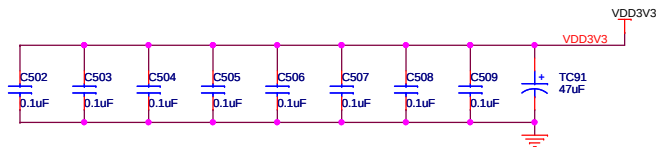
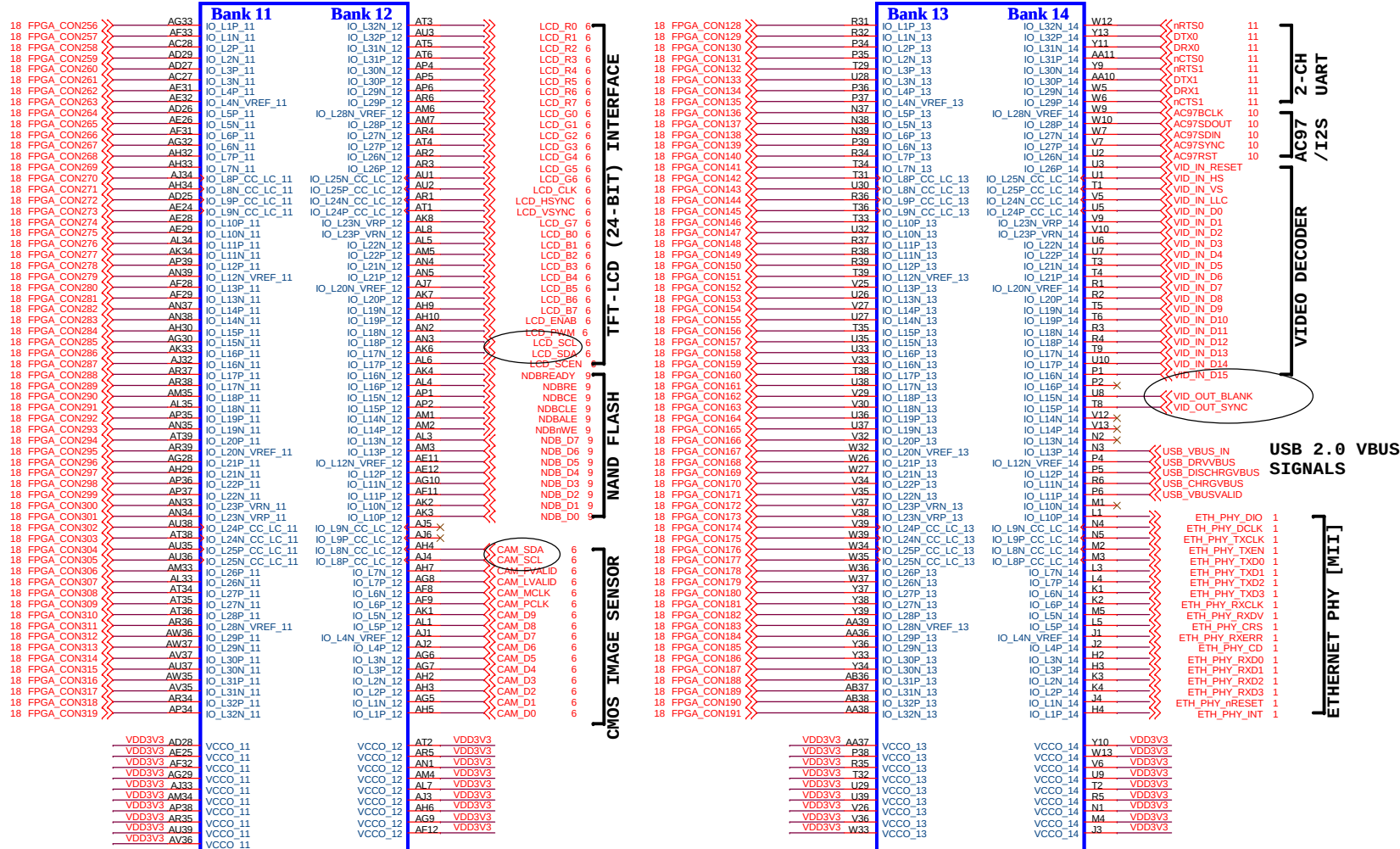




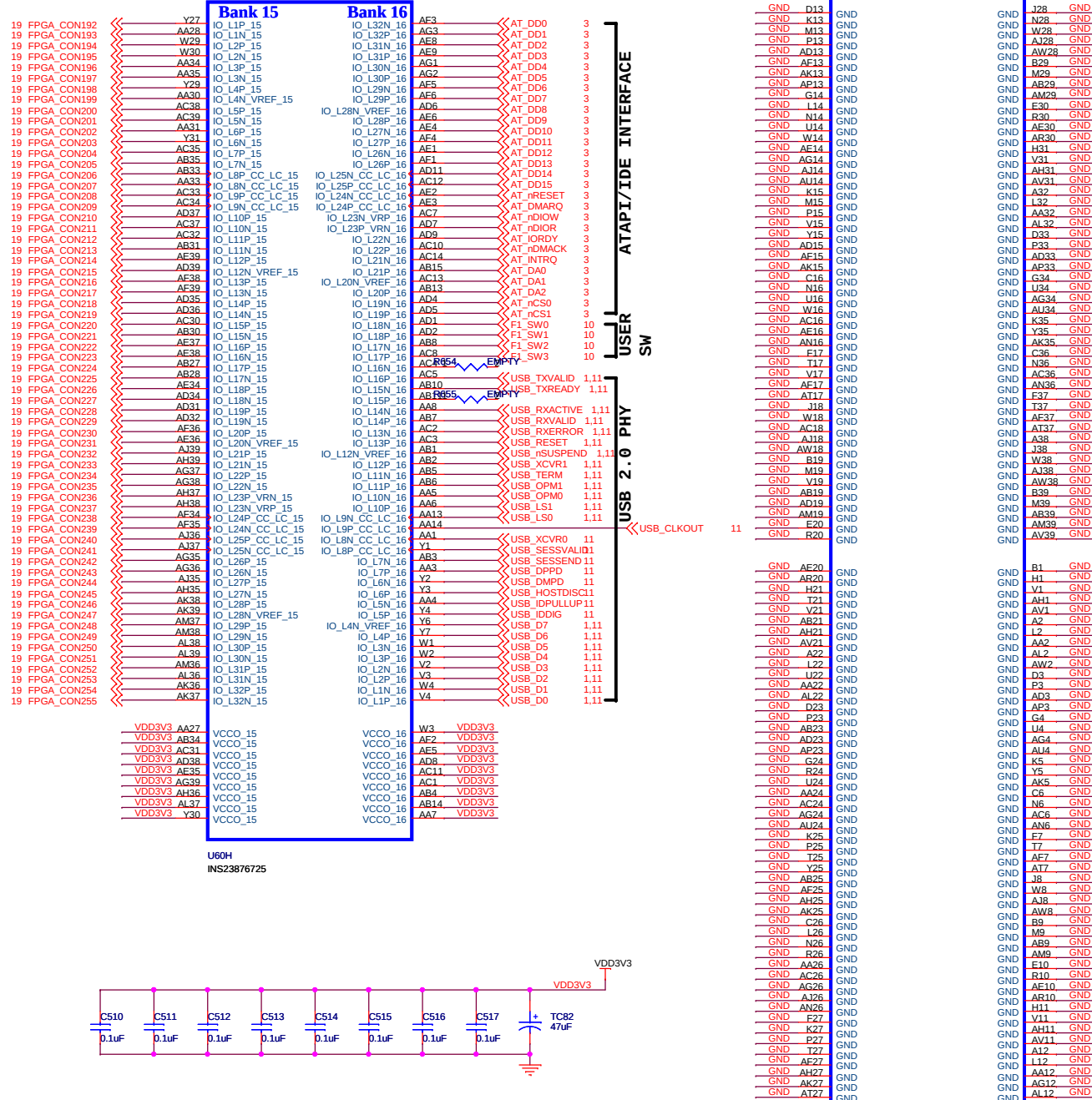
DYNAMIC MEMORY INTERFACE







VIRTEX-4 FPGA POWER GROUND



U60K
INS23878820

